

Margaret E. Verrico

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Department of Astronomy
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Education

University of Illinois Urbana-Champaign, Urbana, IL (2022 - Present)

- Ph.D. Candidate in Astronomy advised by Prof. K. Decker French
- M.S. in Astronomy, August 2024
- CITL Graduate Teaching Certificate 2023, Graduate College Mentoring Certificate 2024

University of Pittsburgh, Pittsburgh, PA (2017 - 2021)

- B.S. – Physics, magna cum laude

Fellowships and Awards

- AAS International Travel Grant, Summer 2026
- UIUC Mr. and Mrs. Hsiang-pai and Wen-hua Chu Department Of Astronomy Excellence in Research Graduate Student Award, Spring 2026
- UIUC Astronomy Department Lewis E. Snyder Memorial Travel Award, Spring 2026
- AAS FAMOUS Grant Awardee, Summer 2025
- Center for Astrophysical Surveys Graduate Fellowship, Fall 2024–Summer 2025
- NSF Graduate Research Fellowship Honorable Mention, Spring 2024
- NASA Illinois Space Grant Graduate Fellowship, Fall 2023–Spring 2024
- NASA Pennsylvania Space Grant, Spring 2019, Summer 2020, Fall 2020, Spring 2021
- University of Pittsburgh full tuition academic scholarship, 2017–2021

Awarded Grants & Telescope Time

As Principal Investigator:

- *A Census of AGN in Rapidly-Quenching Galaxies at Intermediate Redshift with Roman's High-Latitude Wide-Angle Survey*, Roman Space Telescope Cycle 1 Small Program, **\$150,000**.
- GN-2025B-FT-211: *A Repeat Changing-Look AGN Caught in Real Time*, Gemini North 2025B Fast Turnaround. **1.43 h**.
- 2025.1.00888.S: *Resolving molecular gas in the nuclei around changing-look AGN*, ALMA Cycle 12. **20h**.

Co-I Summary: 8 ALMA observing proposals totalling **159.3 hours**; 1 JWST observing proposal totalling **4.7 hours**; 1 JWST archival proposal totalling **\$140,000**; and 1 Roman funding proposal totalling **\$150,000**.

Publications

Summary: 28 total publications with h-index 8. 24 refereed/submitted, 3 research notes, 1 white paper. 3 first-author papers, 2 with significant contribution, and 23 with contribution.

Major Author (* Denotes work led by a student I mentored.)

1. *Identifying Changing-Look AGN Transitions in Light Curve Data with the Zwicky Transient Facility*, **Verrico, M. E.**; French, K. D.; Baldassare, V. F.; et al., 2026, ApJ, 1004, 121
2. *Modeling Star Formation Histories of Changing-Look AGN Host Galaxies with Prospector*, **Verrico, M. E.**; French, K. D.; Suess, K. A.; Agrawal, T., et al., 2025, ApJ, 989, 101
3. **Morphologies of CLAGN Host Galaxies in Galaxy Zoo*, Agrawal, T., **Verrico, M. E.**, French, K. D., 2025, RNAAS, 9, 245
4. *Merger Signatures are Common, but not Universal, in Massive, Recently Quenched Galaxies at $z \sim 0.7$* , **Verrico, M. E.**, Setton, D. J., Bezanson, R., et al., 2023, ApJ, 949, 5
5. *The Compact Structures of Massive $z \sim 0.7$ Post-starburst Galaxies in the SQuIGGLE Sample*, Setton, D. J., **Verrico, M.**, Bezanson, R., et al., 2022, ApJ, 931, 51

Selected Other Publications

6. *The Delay Time Distribution of Quasi-Periodic Eruptions*, Shepherd, M., French, K. D., Hinkle, J.T., et al., ApJL in press, 2026
7. *SQuIGGLE: Ubiquitous Cool Gas Outflows with Large Mass-Loading Factors in $z \sim 0.7$ Post-starburst Galaxies*, Luo, Y., Spilker, J. S., Setton, D. J., et al. submitted to ApJ 2026
8. *Molecular Gas Excitation in $z \sim 0.7$ Gas-Rich Post-starburst Galaxies from SQuIGGLE*, D'Onofrio, V. R., Spilker, J. S., Bezanson, R., et al. 2026, ApJ, 1003, 52
9. *FEADME: Fast Elliptical Accretion Disk Modeling Engine*, Earl, N., French, K.D., Hinkle, J. T. et al. submitted to ApJ 2026, arXiv:2512.10228
10. *The Delay Time Distribution of Tidal Disruption Events*, Shepherd, M., French, K. D., Stone, N., et al. 2026, ApJ, 1005, 63
11. *A Multiwavelength Evaluation of AGN in the Post-Starburst Phase*, Luo, Y., Rowlands, K., Alatalo, K., et al. 2026, ApJ, 1000, 24
12. *Selecting Post-Starburst Galaxies Based on Star Formation History*, Starecheski, S., French, K. D., Villanueva, V., et al. 2026, ApJ, 1000, 110
13. *SQuIGGLE: Buried star formation cannot explain the rapidly fading CO(2-1) luminosity in massive, $z \sim 0.7$ post-starburst galaxies*, Setton, D. J., Spilker, J. S., Bezanson, R., et al. 2025, ApJ, 170, 351
14. *Meet the Neighbors: Gas Rich "Buddy Galaxies" are Common Around Recently Quenched Massive Galaxies in the SQuIGGLE Survey*, Kumar, A., Setton, D. J., Bezanson, R., et al. 2025, RNAAS, 9, 9
15. *Radio Variability in Recently-Quenched Galaxies: The Impact of TDE or AGN Driven Outflows*, French, K. D., Nyland, K., Patil, P., et al. 2025, ApJ, 992, 123
16. *Unusually High Gas-to-Dust Ratios Observed in High-Redshift Quiescent Galaxies*, Spilker, J. S., Whitaker, K. E., Narayanan, D., et al. 2025, ApJL, 993, L40
17. *Cold gas in a post-starburst pair at $z \sim 1.4$: major mergers as a pathway to quenching in the HeavyMetal survey*, Suess, K. A., Beverage, A. G., Kriek, M., et al. 2025, ApJ, 993, 158
18. *AT 2020nov: Evidence for Disk Reprocessing in a Rare Tidal Disruption Event*, Earl, N., French, K. D., Ramirez-Ruiz, E., et al. 2025, ApJ, 983, 28
19. *A Large-Scale Search for Photometrically Variable AGNs in Dwarf Galaxies Using the Young Supernova Experiment*, Messick, A., Baldassare, V., Jones, D. O., et al. 2025, ApJ, 985, 223
20. *Shocked P0ststarburst Galaxy Survey. IV. Outflows in Shocked Poststarburst Galaxies Are Not Responsible for Quenching*, Fodor, A., Tomko, T., Braun, M., et al. 2025, ApJ, 979, 94
21. *Modeling Star Formation Histories of Post Starburst Galaxies with BAGPIPES*, Crocker, D. L., French, K. D., Tripathi, A., & Verrico, M. E., 2023, RNAAS, 7, 183
22. *Fading AGNs in Poststarburst Galaxies*, French, K. D., Earl, N., Novack, A. B., et al. 2023, ApJ, 950, 153
23. *Star Formation Suppression by Tidal Removal of Cold Molecular Gas from an Intermediate-redshift Massive Post-starburst Galaxy*, Spilker, J. S., Suess, K. A., Setton, D. J. et al., 2022, ApJ, 936, L11
24. *Now You See It, Now You Don't: Star Formation Truncation Precedes the Loss of Molecular Gas by 100 Myr in Massive Poststarburst Galaxies at $z \sim 0.6$* , Bezanson, R., Spilker, J. S., Suess, K. A., et al. 2022, ApJ, 925, 153
25. *SQuIGGLE: Studying Quenching in Intermediate- z Galaxies-Gas, Angular Momentum, and Evolution*, Suess, K. A., Kriek, M., Bezanson, R., et al. 2022, ApJ, 926, 89

Talks & Presentations

- *Identifying Changing-Look AGN Transitions in Light Curve Data with the Zwicky Transient Facility*
 1. Talk, Tidal Disruption Events and Nuclear Transients in Crete Vol. 2, Fall 2026, Ibis Styles Heraklion Central Hotel, Heraklion, Crete, Greece
 2. Talk, Galread, Spring 2026, Princeton University, NJ, USA
 3. Talk, Journal Club, Spring 2026, Rutgers University, NJ, USA
 4. Talk, Monday Science Seminar, Spring 2026, University of Wisconsin-Madison, Madison, WI, USA
 5. Talk, Center for AstroPhysical Surveys Seminar, Summer 2025, University of Illinois Urbana-Champaign, Urbana-Champaign, IL, USA
- *A new sample of post-starburst quasars at intermediate redshift from the DESI Early Data Release*
 1. Talk, End of Star Formation conference, Spring 2026, University of Illinois Urbana-Champaign, Urbana-Champaign, IL, USA
- *Modeling Star Formation Histories of Changing-Look AGN Host Galaxies with Prospector*
 1. Talk, X-ray Quasi-Periodic Eruptions & Repeating Nuclear Transients, Summer 2025, European Space Astronomy Centre (ESAC), Madrid, Spain
 2. Poster, European Astronomical Society Annual Meeting, Summer 2025, University College Cork, Cork, Ireland
 3. Poster, Transients From Space Workshop, Spring 2025, Space Telescope Science Institute, Baltimore, MD
 4. Talk, Tidal Disruption Events and Nuclear Transients: Entering the Data-Rich Era, Fall 2024, Aquila Atlantis hotel, Heraklion, Crete, Greece
 5. Flash Talk, Young Astronomers on Galactic Nuclei, Fall 2023, Istituto di Astrofisica Spaziale e Fisica Cosmica, Palermo, Italy
- *Constraining Changing-Look AGN Origins with Host Galaxy Studies*
 1. Talk, 246th Meeting of the AAS, Summer 2025, Anchorage, AK
- *Merger Signatures are Common, but not Universal, in Massive, Recently-Quenched Galaxies at $z \sim 0.7$*
 1. Poster, European Astronomical Society Annual Meeting, Summer 2025, University College Cork, Cork, Ireland
 2. Talk, Journal Club, Fall 2022, University of Illinois Urbana-Champaign, Champaign, IL, USA

Mentoring

Research Mentoring

- Tanay Agrawal, undergraduate Astronomy + Computer Science major, University of Illinois Urbana-Champaign; Spring 2024–Present. Published research note & *Finding tidal tails and AGN in Legacy Imaging of Post-Starburst Galaxies*, Agrawal, T., Verrico, M. E., French, K. D. in prep.
- Jonah Miller, undergraduate Astrophysics/Astronomy + Data Science major, University of Illinois Urbana-Champaign; Summer 2026–Present.
- Aayan Verma, undergraduate Astrophysics major, University of Illinois Urbana-Champaign; Fall 2025–Spring 2026. *Extreme MgII Emission in DESI Post-starbursts: A Signature of Fading Quasars?* Verrico, M. E., Verma, A., French, K. D., et al. in prep.

Career Mentoring through UIUC's Society for Equity in Astronomy

- Adwitiya Sarkar, undergraduate Astrophysics major, University of Illinois Urbana-Champaign; Fall 2022–Spring 2026. Graduated Spring 2026, first-year Ph.D. student at Yale University.
- Zoe Surles, undergraduate Astronomy major, University of Illinois Urbana-Champaign; Fall 2022–Spring 2024. Awarded the Layla S. Ryan Scholarship for outreach and service. Graduated Spring 2025.

Teaching

- Guest lecturer, ASTR 406: *Galaxies and the Universe*, Spring 2026
- Counselor and Activity Lead, UIUC Girls' Astronomy Summer Camp, Summer 2023–Present
- Teaching Assistant, ASTR 121: *Solar Systems and Worlds Beyond*, Spring 2023
- Teaching Assistant, ASTR 122: *Stars and Galaxies*, Fall 2022
- Science Instructor, People's Park Summer Camp, Summer 2022
- Undergraduate Teaching Assistant, *Basic Physics for Science and Engineering I*, Fall 2018

Service & Outreach

- University of Illinois Department of Astronomy graduate admissions committee, Fall 2026–Spring 2027
- Reviewer for The Astrophysical Journal, Co-reviewer for Science Advances, Spring 2025–Present
- First year graduate student mentoring program organizer, Fall 2025–Present
- [Astrobites contributor](#) and Education Committee member, Spring 2025–Present
- Science Olympiad Division B State & Regional Event Coordinator, Spring 2023–Present
- End of Star Formation conference Local Organizing Committee, Spring 2026
- "Stargazing in Illinois" talk, Mahomet Public Library, Fall 2025
- "Galaxy Evolution: How Do Galaxies Stop Forming Stars?" talk, DASH Astronomy, Fall 2025
- ["Galaxies: From Elliptical to Spiral."](#) public talk, Martz-Kohl Observatory, Spring 2025
- "When Stars Explode!" first grade classroom Q&A, Dr. Howard Elementary, Spring 2025
- Undergraduate Research Apprenticeship Program Mentor, Spring 2024
- UIUC Girls' Astronomy Summer Camp Director, Summer 2023–Summer 2024
- UIUC Society for Equity in Astronomy Chair, Fall 2023–Summer 2024
- UIUC Society for Equity in Astronomy mentor, Fall 2022–Present